

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – RAG-BASED MINI ASSIGNMENT (DOCUMENT QA / AI AGENT DEMO)

Course Code:	CSA65	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO3 – Precision (BL2, BL3)	Assessment:	Assessment Tool 2 – RAG-based Mini Assignment (Document QA / AI Agent Demo)
Weightage:	20% of CIA	Total Marks:	2 * 50 = 100 (1 Question1)
CO3 Statement:	<i>Develop intelligent applications using Retrieval-Augmented Generation (RAG), vector databases, and introductory AI agent concepts.(BL3, BL4)</i>		
Student Name:	S.Sirivally	Reg. No.:	192472371
Section / Batch:	2024 / CSE-AI	Date:	11-08-26

Instructions to Students

- Answer 2 questions. Each question carries 50 marks.Total: 100 Marks.
- Implement the solution using **Python**.
- Use an appropriate LLM such as **OpenAI, Gemini, or Hugging Face**.
- Use **embeddings and semantic search** where applicable.
- Use **FAISS or ChromaDB** as the vector database for RAG tasks.
- Demonstrate the complete pipeline:
Document → Chunking → Embedding → Retrieval → Generation
- Demonstrate the system with multiple queries.
- Handle irrelevant, incomplete, or unsupported queries appropriately.
- Display retrieved context/source wherever applicable.
- Explain the system architecture.
- Provide a working end-to-end demonstration.

Code of Conduct

I __S.Sirivally / 192472371__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: __Sirivally.S__

SECTION A – RAG-BASED MINI ASSIGNMENT (DOCUMENT QA / AI AGENT DEMO)

13. Top-K Retrieval Demonstration: Develop a RAG system where the user can select Top-1, Top-3, or Top-5 retrieval and compare the retrieved context.

TOP-K RETRIEVAL DEMONSTRATION

Problem Statement

Develop a Retrieval-Augmented Generation (RAG) system in which the user can select Top-1, Top-3, or Top-5 retrieval and compare the retrieved context for a given query.

The system should process a document, divide it into chunks, generate embeddings, store the embeddings in a FAISS vector database, perform semantic search, retrieve the selected number of relevant chunks, and use the retrieved context to generate an answer using a Large Language Model.

Aim

To implement a Python-based Top-K Retrieval RAG system using Sentence Transformer embeddings, FAISS vector search, and the Hugging Face LLM API, and to compare the retrieved context for Top-1, Top-3, and Top-5 retrieval.

Objectives

1. To implement a complete RAG pipeline using Python.
2. To divide a source document into smaller chunks.
3. To generate vector embeddings using a Sentence Transformer model.
4. To store the embeddings in a FAISS vector database.
5. To perform semantic similarity search.
6. To allow the user to select Top-1, Top-3, or Top-5 retrieval.
7. To display the retrieved context and similarity information.
8. To pass the retrieved context to a Hugging Face LLM.
9. To generate an answer based only on the retrieved information.
10. To handle invalid and irrelevant queries appropriately.

Technologies Used

Technology	Purpose
Python	Main programming language
Hugging Face	LLM platform
openai/gpt-oss-120b	Language model
Sentence Transformers	Embedding generation
all-MiniLM-L6-v2	Embedding model
FAISS	Vector database and similarity search
huggingface_hub	Hugging Face API integration
Python IDLE	Program development and execution

Dataset / Source Document

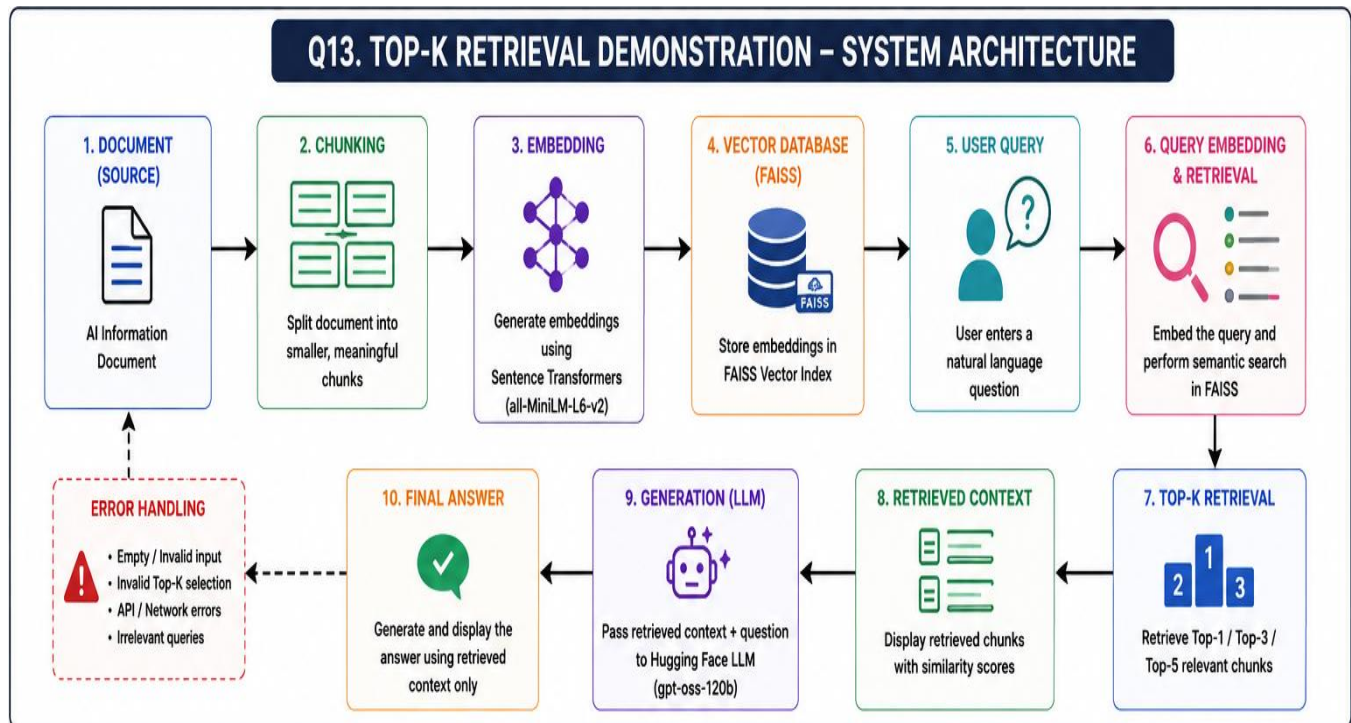
For this experiment, a small document containing information about **Artificial Intelligence** was created.

The document includes information about:

- Artificial Intelligence
- Machine Learning
- AI in Healthcare
- AI in Education
- AI in Banking
- Natural Language Processing
- Computer Vision
- Generative AI

The document is intentionally divided into multiple paragraphs so that each paragraph can be treated as an individual retrieval chunk.

System Architecture



System Architecture of the Top-K Retrieval RAG System

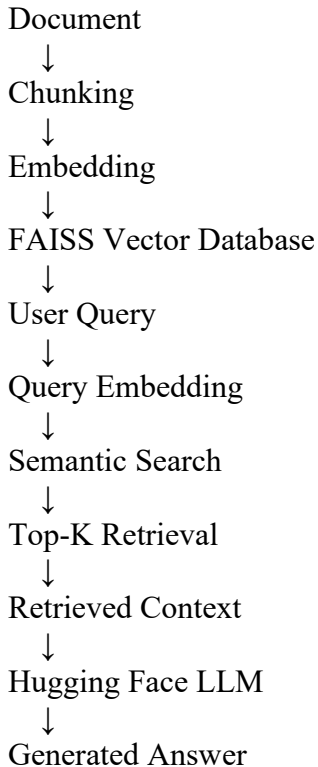
Architecture Explanation

The system consists of the following stages:

1. **Document:** The source document contains information about Artificial Intelligence.
2. **Chunking:** The document is divided into smaller meaningful text chunks.
3. **Embedding:** Each chunk is converted into a numerical vector using Sentence Transformers.
4. **FAISS Vector Database:** The generated vectors are stored in FAISS.
5. **User Query:** The user enters a natural-language question.
6. **Query Embedding:** The query is converted into the same vector representation.
7. **Semantic Retrieval:** FAISS compares the query vector with stored document vectors.
8. **Top-K Selection:** The system retrieves Top-1, Top-3, or Top-5 relevant chunks.
9. **Retrieved Context:** The selected chunks are displayed to the user.
10. **Generation:** The retrieved context is provided to the Hugging Face LLM.
11. **Final Answer:** The LLM generates a response using the retrieved context.

Complete RAG Pipeline

The implemented system follows the required pipeline:



This demonstrates the complete:

Document → Chunking → Embedding → Retrieval → Generation
pipeline required by the assessment.

Working Principle

First, the source document is divided into smaller chunks.

Each chunk is converted into a numerical vector using the Sentence Transformer model:
sentence-transformers/all-MiniLM-L6-v2

These vectors are stored in a FAISS index.

When the user enters a question, the question is also converted into an embedding. FAISS performs a similarity search between the query embedding and the stored document embeddings.

Depending on the user's selection, the system retrieves:

- Top-1 most relevant chunk
- Top-3 most relevant chunks
- Top-5 most relevant chunks

The retrieved chunks are displayed and then passed to the Hugging Face language model. The LLM generates an answer using only the retrieved context.

Algorithm – Top-K Retrieval RAG

Step 1: Start the program.

Step 2: Load the source document.

Step 3: Divide the document into smaller chunks.

Step 4: Load the Sentence Transformer embedding model.
Step 5: Generate embeddings for all document chunks.
Step 6: Create a FAISS vector index.
Step 7: Store the document embeddings in FAISS.
Step 8: Accept a question from the user.
Step 9: Ask the user to select Top-1, Top-3, or Top-5.
Step 10: Generate an embedding for the user query.
Step 11: Perform semantic similarity search using FAISS.
Step 12: Retrieve the selected Top-K chunks.
Step 13: Display the retrieved context and similarity information.
Step 14: Send the retrieved context and question to the Hugging Face LLM.
Step 15: Generate the final answer.
Step 16: Display the generated answer.
Step 17: Handle invalid input and API errors.
Step 18: Stop the program.

Implementation

The program is implemented using Python.

The main functions include:

- `create_chunks()`
- `create_vector_database()`
- `retrieve_documents()`
- `generate_answer()`
- `main()`

The modular structure separates document processing, vector database creation, retrieval, generation, and user interaction.

The program uses:

SentenceTransformer

+

FAISS

+

Hugging Face InferenceClient

to implement the complete RAG pipeline.

The complete source code is attached separately with the submission.

Test Case 1 – Top-5 Retrieval

Objective

To retrieve five relevant chunks and observe how increasing K changes the amount of retrieved context.

User Query

What are the applications of AI in healthcare?

Selection

Top-5

Retrieved Context

The system displays five chunks ranked according to semantic similarity.

The first result is expected to be strongly related to healthcare, while additional results may provide broader information about AI, machine learning, education, or other AI applications.

Generated Answer

The LLM generates an answer using the retrieved context.

Output:

```
IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help

Rank 1
Similarity Distance: 0.5699
Context: Artificial intelligence is widely used in healthcare.
AI systems can help doctors analyze medical images, identify
patterns in patient records, and support clinical decisions.

Rank 2
Similarity Distance: 0.7839
Context: AI is also used in banking and finance. It can help detect
fraudulent transactions, analyze financial data, and support
risk assessment.

Rank 3
Similarity Distance: 1.06
Context: Artificial intelligence is a branch of computer science that
focuses on creating systems capable of performing tasks that
normally require human intelligence.

Rank 4
Similarity Distance: 1.0769
Context: Machine learning is a major part of artificial intelligence.
It allows computers to learn patterns from data and improve
their performance without being explicitly programmed for
every task.

Rank 5
Similarity Distance: 1.094
Context: Computer vision enables computers to analyze images and
videos. Applications include facial recognition, object
detection, medical image analysis, and autonomous vehicles.

Generating answer...
Please wait...

=====
GENERATED ANSWER
=====
AI applications in healthcare include:

- Analyzing medical images (e.g., radiology, pathology)
- Identifying patterns in patient records
- Supporting clinical decision-making

Computer vision techniques also enable medical image analysis.
>>>
```

```
CO3_AT2_Q13_TopK_RAG.py - D:/CSA6502-Gen AI/CO3_AT2_Q13_TopK_RAG.py (3.12.10)
File Edit Format Run Options Window Help

focuses on creating systems capable of performing tasks that
normally require human intelligence.

Machine learning is a major part of artificial intelligence.
It allows computers to learn patterns from data and improve
their performance without being explicitly programmed for
every task.

Artificial intelligence is widely used in healthcare.
AI systems can help doctors analyze medical images, identify
patterns in patient records, and support clinical decisions.

In education, artificial intelligence can provide personalized
learning experiences. AI systems can recommend learning
materials based on student performance and can provide
automated feedback.

AI is also used in banking and finance. It can help detect
fraudulent transactions, analyze financial data, and support
risk assessment.

Natural language processing allows computers to understand
and generate human language. It is used in chatbots,
translation systems, text summarization, and virtual assistants.

Computer vision enables computers to analyze images and
videos. Applications include facial recognition, object
detection, medical image analysis, and autonomous vehicles.

Generative AI is a type of artificial intelligence that can
create new content such as text, images, audio, and code.
Large language models are an important technology behind
many generative AI applications.

"""

# =====
# CHUNKING
# =====

def create_chunks(document):

    paragraphs = [
        p.strip()
        for p in document.split("\n\n")
        if p.strip()
    ]
```

```
IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help

===== RESTART: D:/CSA6502-Gen AI/CO3_AT2_Q13_TopK_RAG.py =====
===== Q13 - TOP-K RETRIEVAL RAG SYSTEM =====

Document loaded successfully.
Number of chunks: 8

Loading embedding model...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
o enable higher rate limits and faster downloads.
Loading weights: 0% | 0/103 [00:00<00:00, 20623.13it/s] Loading weights: 100%|
| 103/103 [00:00<00:00, 20623.13it/s]
Embedding model loaded.

Creating FAISS vector database...
FAISS database created.
Number of vectors: 8

Enter your question: What are the applications of AI in healthcare?

Select retrieval size:
1 - Top-1
3 - Top-3
5 - Top-5
Enter 1, 3, or 5: 5

Performing semantic search...
Retrieving Top- 5 documents...

===== RETRIEVED CONTEXT =====

Rank 1
Similarity Distance: 0.5699
Context: Artificial intelligence is widely used in healthcare.
AI systems can help doctors analyze medical images, identify
patterns in patient records, and support clinical decisions.

Rank 2
Similarity Distance: 0.7839
Context: AI is also used in banking and finance. It can help detect
fraudulent transactions, analyze financial data, and support
risk assessment.
```

```
CO3_AT2_Q13_TopK_RAG.py - D:/CSA6502-Gen AI/CO3_AT2_Q13_TopK_RAG.py (3.12.10)
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"""

# =====
# CHUNKING
# =====

def create_chunks(document):

    paragraphs = [
        p.strip()
        for p in document.split("\n\n")
        if p.strip()
    ]
```

Test Case 2 – Top-1 Retrieval

Objective

To retrieve only the single most relevant document chunk for a user query.

User Query

What are the applications of AI in healthcare?

Selection

Top-1

Retrieved Context

The system retrieves the most relevant chunk related to healthcare.

Example:

Rank 1

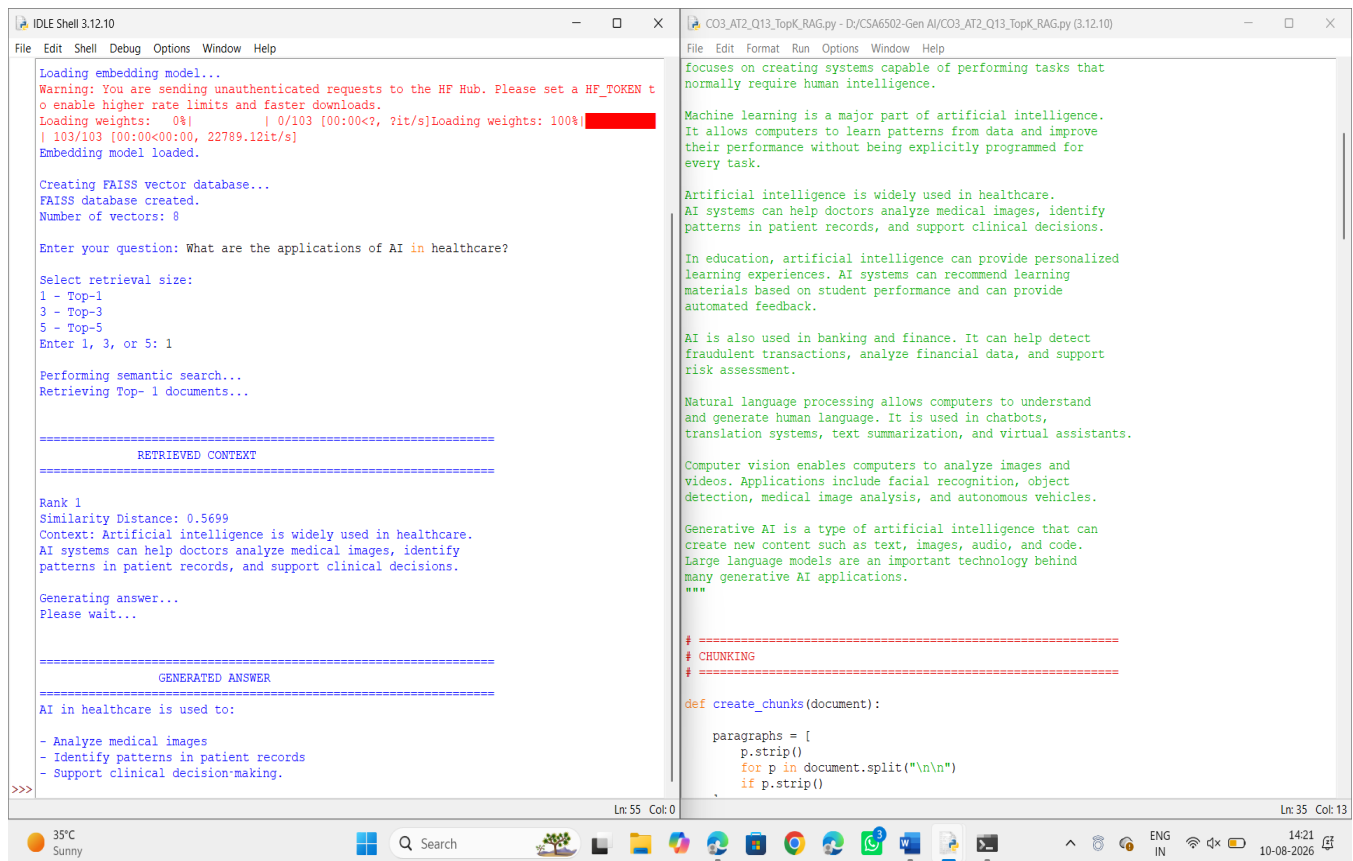
Context:

Artificial intelligence is widely used in healthcare. AI systems can help doctors analyze medical images, identify patterns in patient records, and support clinical decisions.

Generated Answer

Artificial Intelligence is used in healthcare to help analyze medical images, identify patterns in patient records, and support clinical decision-making.

Output:



```

IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Loading embedding model...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
o enable higher rate limits and faster downloads.
Loading weights: 0% | 0/103 [00:00<?, ?it/s]Loading weights: 100% ██████████
| 103/103 [00:00<00:00, 22789.12it/s]
Embedding model loaded.

Creating FAISS vector database...
FAISS database created.
Number of vectors: 8

Enter your question: What are the applications of AI in healthcare?

Select retrieval size:
1 - Top-1
3 - Top-3
5 - Top-5
Enter 1, 3, or 5: 1

Performing semantic search...
Retrieving Top- 1 documents...

=====
RETRIEVED CONTEXT
=====

Rank 1
Similarity Distance: 0.5699
Context: Artificial intelligence is widely used in healthcare.
AI systems can help doctors analyze medical images, identify
patterns in patient records, and support clinical decisions.

Generating answer...
Please wait...

=====
GENERATED ANSWER
=====
AI in healthcare is used to:

- Analyze medical images
- Identify patterns in patient records
- Support clinical decision-making.
>>>

CO3_AT2_Q13_TopK_RAG.py - D:\CSA6502-Gen AI\CO3_AT2_Q13_TopK_RAG.py (3.12.10)
File Edit Format Run Options Window Help
focuses on creating systems capable of performing tasks that
normally require human intelligence.

Machine learning is a major part of artificial intelligence.
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their performance without being explicitly programmed for
every task.

Artificial intelligence is widely used in healthcare.
AI systems can help doctors analyze medical images, identify
patterns in patient records, and support clinical decisions.

In education, artificial intelligence can provide personalized
learning experiences. AI systems can recommend learning
materials based on student performance and can provide
automated feedback.

AI is also used in banking and finance. It can help detect
fraudulent transactions, analyze financial data, and support
risk assessment.

Natural language processing allows computers to understand
and generate human language. It is used in chatbots,
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videos. Applications include facial recognition, object
detection, medical image analysis, and autonomous vehicles.

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create new content such as text, images, audio, and code.
Large language models are an important technology behind
many generative AI applications.
"""

=====
CHUNKING
=====

def create_chunks(document):

    paragraphs = [
        p.strip()
        for p in document.split("\n\n")
        if p.strip()
    ]

```

Test Case 3 – Top-3 Retrieval

Objective

To retrieve the three most relevant chunks and compare the retrieved context with Top-1 retrieval.

User Query

What are the applications of AI in healthcare?

Selection

Top-3

Retrieved Context

The system displays three retrieved chunks ranked according to semantic similarity.

Example:

Rank 1

Artificial intelligence is widely used in healthcare...

Rank 2

Machine learning is a major part of artificial intelligence...

Rank 3

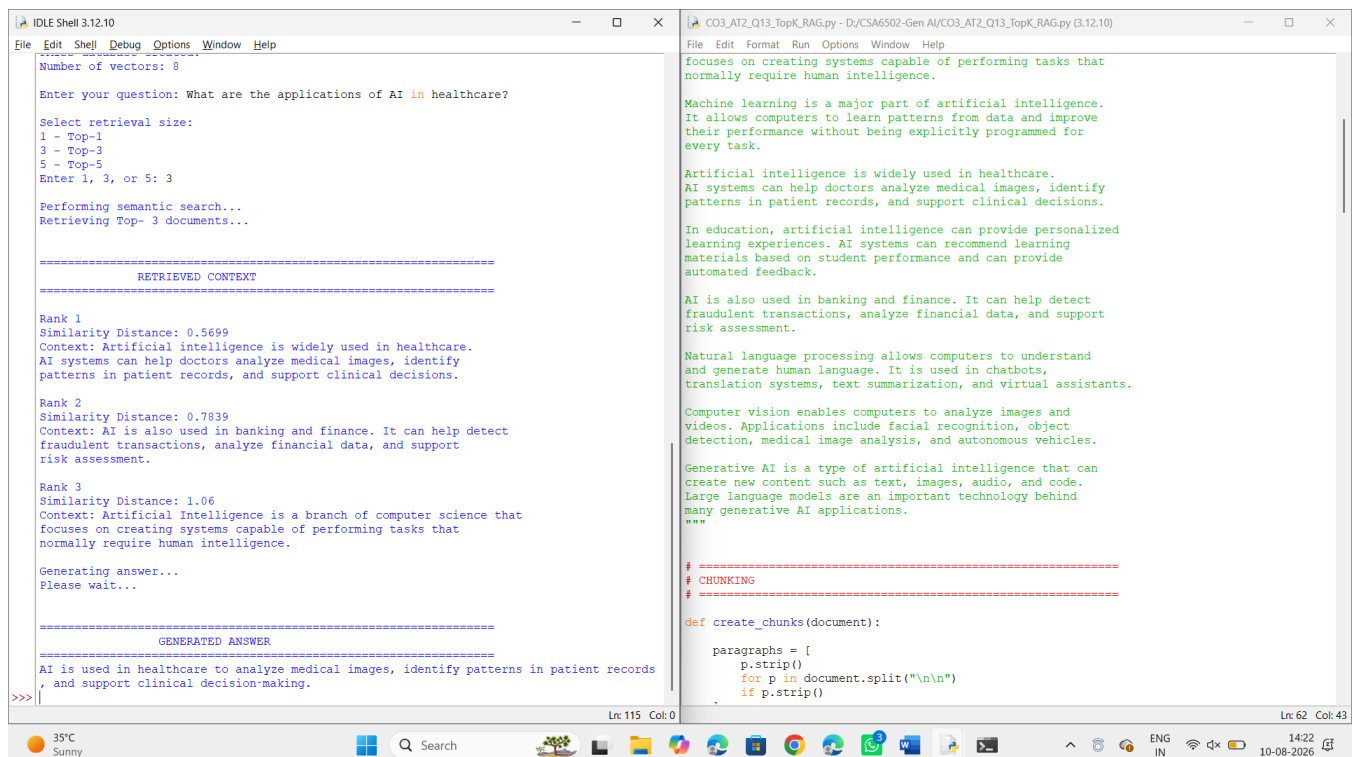
In education, artificial intelligence can provide personalized learning experiences...

The exact retrieved ranking depends on the embedding similarity calculated by FAISS.

Generated Answer

Artificial Intelligence can support healthcare by analyzing medical images, identifying patterns in patient data, and assisting clinical decisions.

Output:



```

IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Number of vectors: 8
Enter your question: What are the applications of AI in healthcare?
Select retrieval size:
1 - Top-1
3 - Top-3
5 - Top-5
Enter 1, 3, or 5: 3
Performing semantic search...
Retrieving Top- 3 documents...

=====
RETRIEVED CONTEXT
=====

Rank 1
Similarity Distance: 0.5699
Context: Artificial intelligence is widely used in healthcare.
AI systems can help doctors analyze medical images, identify
patterns in patient records, and support clinical decisions.

Rank 2
Similarity Distance: 0.7839
Context: AI is also used in banking and finance. It can help detect
fraudulent transactions, analyze financial data, and support
risk assessment.

Rank 3
Similarity Distance: 1.06
Context: Artificial intelligence is a branch of computer science that
focuses on creating systems capable of performing tasks that
normally require human intelligence.

Generating answer...
Please wait...

=====
GENERATED ANSWER
=====
AI is used in healthcare to analyze medical images, identify patterns in patient records
, and support clinical decision-making.
>>>

CO3_AT2_Q13_TopK_RAG.py - D:\CSA6502-Gen AI\CO3_AT2_Q13_TopK_RAG.py (3.12.10)
File Edit Format Run Options Window Help
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patterns in patient records, and support clinical decisions.

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automated feedback.

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Natural language processing allows computers to understand
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videos. Applications include facial recognition, object
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Generative AI is a type of artificial intelligence that can
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Large language models are an important technology behind
many generative AI applications.
"""

# =====
# CHUNKING
# =====

def create_chunks(document):

    paragraphs = [
        p.strip()
        for p in document.split("\n\n")
        if p.strip()
    ]

```


Comparison of Top-K Retrieval

Retrieval Method	Number of Chunks	Context	Observation
Top-1	1	Highly focused	Most concise context
Top-3	3	More supporting information	Balanced retrieval
Top-5	5	Broader context	More information but may include less relevant chunks

Interpretation

Top-1 provides the most focused context and is useful when one highly relevant chunk is sufficient.

Top-3 provides additional supporting information while maintaining good relevance. It provides a useful balance between context size and relevance.

Top-5 provides a broader context. However, some retrieved chunks may be less directly related to the query because more results are included.

Therefore, increasing K increases the amount of contextual information available to the LLM, but it may also introduce less relevant information.

Test Case 4 – Natural Language Processing Query

User Query

How is natural language processing used?

Selection

Top-3

Retrieved Context

The system retrieves the chunk related to Natural Language Processing and other semantically related chunks.

Example:

Natural language processing allows computers to understand and generate human language. It is used in chatbots, translation systems, text summarization, and virtual assistants.

Generated Answer

Natural language processing enables computers to understand and generate human language. It is used in chatbots, translation, text summarization and virtual assistants.

OUTPUT:

```
=====
RESTART: D:/CSA6502-Gen AI/CO3_AT2_Q13_TopK_RAG.py =====
=====
Q13 - TOP-K RETRIEVAL RAG SYSTEM
=====

Document loaded successfully.
Number of chunks: 8

Loading embedding model...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 0% | 0/103 [00:00<00:00, 7308.14it/s] Loading weights: 100% [REDACTED]
Embedding model loaded.

Creating FAISS vector database...
FAISS database created.
Number of vectors: 8

Enter your question: How is natural language processing used?

Select retrieval size:
1 - Top-1
3 - Top-3
5 - Top-5
Enter 1, 3, or 5: 3

Performing semantic search...
Retrieving Top- 3 documents...

=====
RETRIEVED CONTEXT
=====

Rank 1
Similarity Distance: 0.4567
Context: Natural language processing allows computers to understand and generate human language. It is used in chatbots, translation systems, text summarization, and virtual assistants.

Rank 2
Similarity Distance: 1.2728
Context: Generative AI is a type of artificial intelligence that can create new content such as text, images, audio, and code. Large language models are an important technology behind many generative AI applications.

=====
CO3_AT2_Q13_TopK_RAG.py - D:/CSA6502-Gen AI/CO3_AT2_Q13_TopK_RAG.py (3.12.10)
File Edit Format Run Options Window Help

focuses on creating systems capable of performing tasks that normally require human intelligence.

Machine learning is a major part of artificial intelligence. It allows computers to learn patterns from data and improve their performance without being explicitly programmed for every task.

Artificial intelligence is widely used in healthcare. AI systems can help doctors analyze medical images, identify patterns in patient records, and support clinical decisions.

In education, artificial intelligence can provide personalized learning experiences. AI systems can recommend learning materials based on student performance and can provide automated feedback.

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"""
# =====
# CHUNKING
# =====
def create_chunks(document):
    paragraphs = [
        p.strip()
        for p in document.split("\n\n")
        if p.strip()
    ]
    """
```

```
=====
RETRIEVED CONTEXT
=====

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Similarity Distance: 0.4567
Context: Natural language processing allows computers to understand and generate human language. It is used in chatbots, translation systems, text summarization, and virtual assistants.

Rank 2
Similarity Distance: 1.2728
Context: Generative AI is a type of artificial intelligence that can create new content such as text, images, audio, and code. Large language models are an important technology behind many generative AI applications.

Rank 3
Similarity Distance: 1.3288
Context: Machine learning is a major part of artificial intelligence. It allows computers to learn patterns from data and improve their performance without being explicitly programmed for every task.

Generating answer...
Please wait...

=====
GENERATED ANSWER
=====

Natural language processing is used to enable computers to understand and generate human language, powering applications such as chatbots, translation systems, text summarization tools, and virtual assistants.

=====
CO3_AT2_Q13_TopK_RAG.py - D:/CSA6502-Gen AI/CO3_AT2_Q13_TopK_RAG.py (3.12.10)
File Edit Format Run Options Window Help

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"""
# =====
# CHUNKING
# =====
def create_chunks(document):
    paragraphs = [
        p.strip()
        for p in document.split("\n\n")
        if p.strip()
    ]
    """
```

Test Case 5 – Irrelevant Query

User Query

What is the recipe for pizza?

Expected Behaviour

The query is unrelated to the source document.

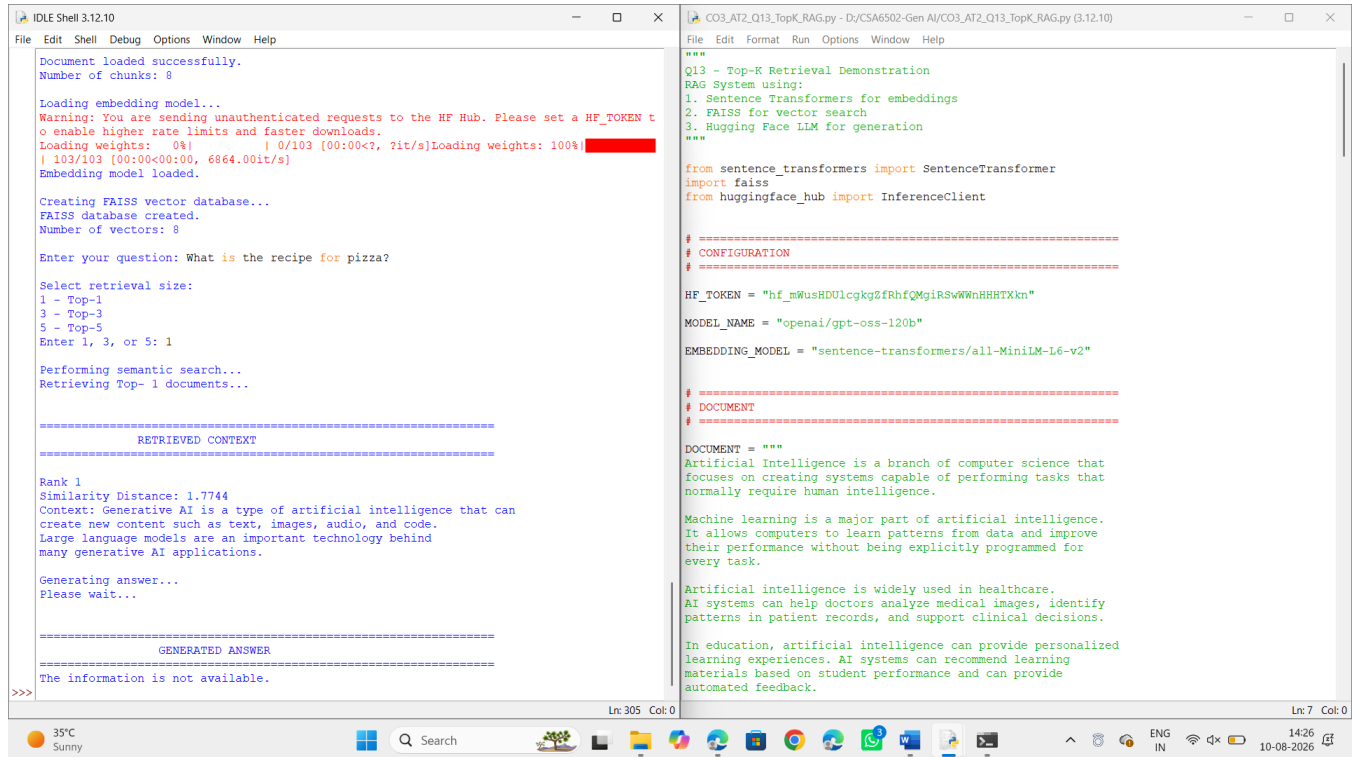
The system still performs semantic retrieval, but the LLM is instructed to answer only from the retrieved context and not invent information.

Expected Output

The requested information is not available in the provided context.

The exact wording may vary depending on the LLM response.

OUTPUT:



```

IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Document loaded successfully.
Number of chunks: 8
Loading embedding model...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 0% | 0/103 [00:00<?, ?it/s]Loading weights: 100% [redacted]
Embedding model loaded.

Creating FAISS vector database...
FAISS database created.
Number of vectors: 8

Enter your question: What is the recipe for pizza?

Select retrieval size:
1 - Top-1
3 - Top-3
5 - Top-5
Enter 1, 3, or 5: 1

Performing semantic search...
Retrieving Top- 1 documents...

=====
RETRIEVED CONTEXT
=====

Rank 1
Similarity Distance: 1.7744
Context: Generative AI is a type of artificial intelligence that can create new content such as text, images, audio, and code. Large language models are an important technology behind many generative AI applications.

Generating answer...
Please wait...

=====
GENERATED ANSWER
=====

>>> The information is not available.

CO3_AT2_Q13_TopK_RAG.py - D:/CSA6502-Gen AI/CO3_AT2_Q13_TopK_RAG.py (3.12.10)
File Edit Format Run Options Window Help
"""
Q13 - Top-K Retrieval Demonstration
RAG System using:
1. Sentence Transformers for embeddings
2. FAISS for vector search
3. Hugging Face LLM for generation
"""

from sentence_transformers import SentenceTransformer
import faiss
from huggingface_hub import InferenceClient

# CONFIGURATION
HF_TOKEN = "hf_mWuSHDulcgkgZfRhQMgiRSwWnRHHTXkn"
MODEL_NAME = "openai/gpt-oss-120b"
EMBEDDING_MODEL = "sentence-transformers/all-MiniLM-L6-v2"

# DOCUMENT
DOCUMENT = """
Artificial Intelligence is a branch of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence.

Machine learning is a major part of artificial intelligence. It allows computers to learn patterns from data and improve their performance without being explicitly programmed for every task.

Artificial intelligence is widely used in healthcare. AI systems can help doctors analyze medical images, identify patterns in patient records, and support clinical decisions.

In education, artificial intelligence can provide personalized learning experiences. AI systems can recommend learning materials based on student performance and can provide automated feedback.
"""

```

Error Handling

The system handles the following cases:

Empty Query

[INPUT ERROR]

Question cannot be empty.

Invalid Top-K Selection

If the user enters a value other than 1, 3, or 5:

[INPUT ERROR]

Please select 1, 3, or 5.

API Error

If the Hugging Face API fails, the program displays an API error instead of terminating unexpectedly.

Result

The Top-K Retrieval RAG system was successfully implemented using Python, Sentence Transformer embeddings, FAISS, and the Hugging Face LLM API.

The system successfully demonstrates Top-1, Top-3, and Top-5 semantic retrieval. The retrieved context is displayed before generation, allowing the effect of different K values to be compared.

The system also handles irrelevant and invalid queries appropriately.

18. Library Information RAG System: Create a RAG system using library rules. Answer questions about borrowing, renewal, fines, timings, and membership.

LIBRARY INFORMATION RAG SYSTEM

Problem Statement

Develop a Retrieval-Augmented Generation system using a library rules document. The system should answer questions related to library borrowing, renewal, fines, timings, membership, reference books, lost books, and other library services.

The system should retrieve relevant information from the library document before generating the answer and should not provide unsupported information for irrelevant questions.

Aim

To develop a Library Information RAG System using Python, Sentence Transformer embeddings, FAISS vector search, and a Hugging Face LLM to answer questions based on a predefined library rules document.

Objectives

1. To create a library rules knowledge document.
2. To divide the document into meaningful chunks.
3. To generate embeddings for the chunks.
4. To store the embeddings in FAISS.
5. To perform semantic search for user questions.
6. To retrieve relevant library information.
7. To display the retrieved context/source.
8. To generate answers using the Hugging Face LLM.
9. To demonstrate the system with multiple library queries.
10. To handle irrelevant and unsupported questions appropriately.

Library Knowledge Base

The knowledge base contains rules related to:

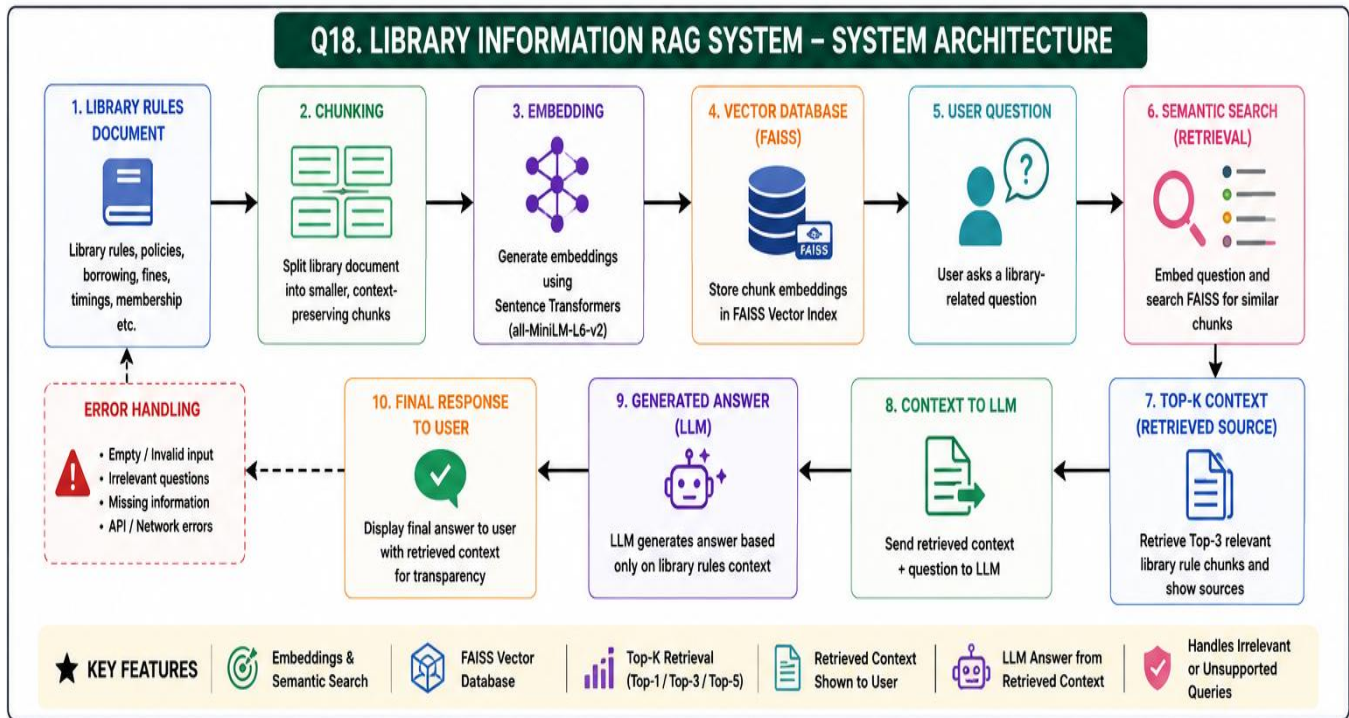
- Library membership
- Borrowing
- Renewal
- Fines
- Library timings
- Reference books
- Lost books
- Damaged books
- Digital resources

Important Library Rules

Category	Rule
Membership	Currently enrolled students can become members

Borrowing	Up to 4 books
Loan period	14 days
Renewal	Once for another 7 days
Fine	Rs. 2 per day per overdue book
Monday-Friday	8:00 AM – 8:00 PM
Saturday	9:00 AM – 5:00 PM
Sunday	Closed
Reference books	Normally cannot be borrowed for home use

System Architecture



System Architecture of the Library Information RAG System

Architecture Explanation

The architecture consists of:

Library Rules Document

↓
Chunking

↓
Embeddings

↓
FAISS Vector Database

↓
User Query

↓
Semantic Retrieval

↓
Retrieved Library Context
↓

Hugging Face LLM



Final Answer

The system ensures that the generated response is based on the information retrieved from the library knowledge base.

Complete RAG Pipeline

The library system follows:

Library Document



Chunking



Embedding



FAISS Vector Database



User Question



Semantic Search



Top-3 Retrieved Context



Hugging Face LLM



Library Answer

This satisfies the required:

Document → Chunking → Embedding → Retrieval → Generation
pipeline.

Working Principle

The library rules are stored as a text document.

The document is divided into individual chunks representing different rules.

Each chunk is converted into an embedding using:

sentence-transformers/all-MiniLM-L6-v2

The embeddings are stored in FAISS.

When a user asks a question, the question is converted into an embedding. FAISS performs semantic similarity search and retrieves the three most relevant chunks.

The retrieved chunks are displayed as the retrieved context/source.

The retrieved information is then sent to the Hugging Face LLM, which generates a concise answer.

The prompt instructs the model not to invent information. If the answer is not available in the library rules, it must clearly indicate that the information is unavailable.

Algorithm – Library Information RAG

Step 1: Start the program.

Step 2: Load the library rules document.

Step 3: Divide the document into chunks.

Step 4: Load the Sentence Transformer embedding model.
Step 5: Generate embeddings for all chunks.
Step 6: Create a FAISS vector database.
Step 7: Store the document embeddings.
Step 8: Accept a library-related question from the user.
Step 9: Convert the question into an embedding.
Step 10: Perform semantic search using FAISS.
Step 11: Retrieve the Top-3 relevant chunks.
Step 12: Display the retrieved context.
Step 13: Send the question and retrieved context to the Hugging Face LLM.
Step 14: Generate the final answer.
Step 15: Display the answer.
Step 16: Handle irrelevant, incomplete, and unsupported questions.

Implementation

The program contains modular functions for:

- Creating document chunks
- Creating the FAISS database
- Retrieving relevant information
- Generating the final answer
- Handling user input

The main RAG components are:

SentenceTransformer



Embeddings



FAISS



Semantic Search



Hugging Face LLM

The complete Python source code is attached separately.

Test Case 1 – Borrowing Rules

User Query

How many books can I borrow?

Retrieved Context

The system retrieves the borrowing rule:

Students can borrow up to 4 books at a time.

Books can normally be borrowed for 14 days.

Generated Output

Students can borrow up to 4 books at a time.

Output:

```
CO3_AT2_Q18_Library_RAG.py - D:/CSA6502-Gen AI/CO3_AT2_Q18_Library_RAG.py (3.12.10)
File Edit Format Run Options Window Help

"""
Q18 - Library Information RAG System

Pipeline:
Document -> Chunking -> Embedding -> FAISS Retrieval
-> Hugging Face LLM -> Answer
"""

from sentence_transformers import SentenceTransformer
import faiss
from huggingface_hub import InferenceClient

# CONFIGURATION
# =====

HF_TOKEN = "hf_mWuSDUlcgkgzFhfgMgIRSwWnHHHTXkn"

MODEL = "openai/gpt-oss-120b"

EMBEDDING_MODEL = "sentence-transformers/all-MiniLM-L6-v2"

# =====

# LIBRARY RULES DOCUMENT
# =====

LIBRARY_RULES = """
Library Membership

All currently enrolled students can become members of the
college library by registering with their student ID.

Borrowing Rules

Students can borrow up to 4 books at a time.
Books can normally be borrowed for 14 days.

Renewal Rules

A borrowed book can be renewed once for another 7 days,
provided that the book has not been reserved by another user.

Fines
"""

IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help

>>> Enter "help" below or click "Help" above for more information.

===== RESTART: D:/CSA6502-Gen AI/CO3_AT2_Q18_Library_RAG.py =====
LIBRARY INFORMATION RAG
=====

Document chunks: 18

Loading embedding model...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
o enable higher rate limits and faster downloads.
Loading weights: 0% | 0/103 [00:00<7, ?it/s]Loading weights: 100% | 
103/103 [00:00<00:00, 7924.23it/s]
FAISS vector database created.

Ask a library question: How many books can I borrow?

Performing semantic search...

===== RETRIEVED CONTEXT =====

Source 1
Distance: 0.5427
Content: Students can borrow up to 4 books at a time.
Books can normally be borrowed for 14 days.

Source 2
Distance: 0.8946
Content: A borrowed book can be renewed once for another 7 days,
provided that the book has not been reserved by another user.

Source 3
Distance: 1.0029
Content: Reference books are available for use inside the library
and cannot normally be borrowed for home use.

===== GENERATED ANSWER =====

You can borrow up to 4 books at a time.
```

Test Case 2 – Renewal

User Query

Can I renew a borrowed book?

Retrieved Context

A borrowed book can be renewed once for another 7 days, provided that the book has not been reserved by another user.

Generated Output

Yes. A borrowed book can be renewed once for another 7 days, provided it has not been reserved by another user.

Output:

```
CO3_AT2_Q18_Library_RAG.py - D:/CSA6502-Gen AI/CO3_AT2_Q18_Library_RAG.py (3.12.10)
File Edit Format Run Options Window Help

"""
Q18 - Library Information RAG System

Pipeline:
Document -> Chunking -> Embedding -> FAISS Retrieval
-> Hugging Face LLM -> Answer
"""

from sentence_transformers import SentenceTransformer
import faiss
from huggingface_hub import InferenceClient

# CONFIGURATION
# =====

HF_TOKEN = "hf_mWuSDUlcgkgzFhfgMgIRSwWnHHHTXkn"

MODEL = "openai/gpt-oss-120b"

EMBEDDING_MODEL = "sentence-transformers/all-MiniLM-L6-v2"

# =====

# LIBRARY RULES DOCUMENT
# =====

LIBRARY_RULES = """
Library Membership

All currently enrolled students can become members of the
college library by registering with their student ID.

Borrowing Rules

Students can borrow up to 4 books at a time.
Books can normally be borrowed for 14 days.

Renewal Rules

A borrowed book can be renewed once for another 7 days,
provided that the book has not been reserved by another user.

Fines
"""

IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help

>>> ===== RESTART: D:/CSA6502-Gen AI/CO3_AT2_Q18_Library_RAG.py =====
LIBRARY INFORMATION RAG
=====

Document chunks: 18

Loading embedding model...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
o enable higher rate limits and faster downloads.
Loading weights: 0% | 0/103 [00:00<7, ?it/s]Loading weights: 100% | 
103/103 [00:00<00:00, 8570.33it/s]
FAISS vector database created.

Ask a library question: Can I renew a borrowed book?

Performing semantic search...

===== RETRIEVED CONTEXT =====

Source 1
Distance: 0.3729
Content: A borrowed book can be renewed once for another 7 days,
provided that the book has not been reserved by another user.

Source 2
Distance: 0.9142
Content: Reference books are available for use inside the library
and cannot normally be borrowed for home use.

Source 3
Distance: 0.9157
Content: Students can borrow up to 4 books at a time.
Books can normally be borrowed for 14 days.

===== GENERATED ANSWER =====

Yes. A borrowed book can be renewed once for an additional 7 days, as long as it hasn't
been reserved by another user.
```


Test Case 3 – Fine

User Query

How much is the fine for an overdue book?

Retrieved Context

A fine of Rs. 2 per day is charged for each overdue book.

Generated Output

The fine for an overdue book is Rs. 2 per day.

```
CO3_AT2_Q18_Library_RAG.py - D:/CSA6502-Gen AI/CO3_AT2_Q18_Library_RAG.py (3.12.10)
File Edit Format Run Options Window Help

"""
Q18 - Library Information RAG System

Pipeline:
Document -> Chunking -> Embedding -> FAISS Retrieval
        -> Hugging Face LLM -> Answer
"""

from sentence_transformers import SentenceTransformer
import faiss
from huggingface_hub import InferenceClient

# =====
# CONFIGURATION
# =====

HF_TOKEN = "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"
MODEL = "openai/gpt-oss-120b"
EMBEDDING_MODEL = "sentence-transformers/all-MiniLM-L6-v2"

# =====
# LIBRARY RULES DOCUMENT
# =====

LIBRARY_RULES = """
Library Membership

All currently enrolled students can become members of the
college library by registering with their student ID.

Borrowing Rules

Students can borrow up to 4 books at a time.
Books can normally be borrowed for 14 days.

Renewal Rules

A borrowed book can be renewed once for another 7 days,
provided that the book has not been reserved by another user.

Fines
"""

IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help

>>>
===== RESTART: D:/CSA6502-Gen AI/CO3_AT2_Q18_Library_RAG.py =====
LIBRARY INFORMATION RAG
=====

Document chunks: 18

Loading embedding model...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
o enable higher rate limits and faster downloads.
Loading weights: 0% | 0/103 [00:00<7, ?it/s]Loading weights: 100%|
| 103/103 [00:00<00:00, 5371.16it/s]
FAISS vector database created.

Ask a library question: How much is the fine for an overdue book?

Performing semantic search...

=====
RETRIEVED CONTEXT
=====

Source 1
Distance: 0.5357
Content: A fine of Rs. 2 per day is charged for each overdue book.
Fines must be cleared before additional borrowing.

Source 2
Distance: 0.9643
Content: Users are responsible for books issued to them.
Damaged books must be reported to library staff.

Source 3
Distance: 0.9676
Content: A lost book must be reported to the library immediately.
The member may be required to replace the book or pay the
applicable replacement cost.

=====
GENERATED ANSWER
=====

The fine is Rs.2 per day for each overdue book.
>>>
```

Test Case 4 – Library Timings

User Query

What are the library timings on Saturday?

Retrieved Context

Saturday: 9:00 AM to 5:00 PM.

Generated Output

The library is open from 9:00 AM to 5:00 PM on Saturday.

Output:

```
CO3_AT2_Q18_Library_RAG.py - D:/CSA6502-Gen AI/CO3_AT2_Q18_Library_RAG.py (3.12.10)
File Edit Format Run Options Window Help

"""
Q18 - Library Information RAG System

Pipeline:
Document -> Chunking -> Embedding -> FAISS Retrieval
-> Hugging Face LLM -> Answer
"""

from sentence_transformers import SentenceTransformer
import faiss
from huggingface_hub import InferenceClient

# CONFIGURATION
# =====
HF_TOKEN = "hf_mWuSHDULcgkgZfRhFqMgIRSwWnRHHTXkn"
MODEL = "openai/gpt-oss-120b"
EMBEDDING_MODEL = "sentence-transformers/all-MiniLM-L6-v2"

# =====
# LIBRARY RULES DOCUMENT
# =====
LIBRARY_RULES = """
Library Membership

All currently enrolled students can become members of the
college library by registering with their student ID.

Borrowing Rules

Students can borrow up to 4 books at a time.
Books can normally be borrowed for 14 days.

Renewal Rules

A borrowed book can be renewed once for another 7 days,
provided that the book has not been reserved by another user.

Fines
"""

IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help

>>> The fine is Rs.2 per day for each overdue book.

===== RESTART: D:/CSA6502-Gen AI/CO3_AT2_Q18_Library_RAG.py =====
LIBRARY INFORMATION RAG

Document chunks: 18
Loading embedding model...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
o enable higher rate limits and faster downloads.
Loading weights: 0% | 0/103 [00:00<?, ?it/s]Loading weights: 100% | ██████████
| 103/103 [00:00<00:00, 14676.36it/s]
FAISS vector database created.

Ask a library question: What are the library timings on Saturday?

Performing semantic search...

===== RETRIEVED CONTEXT =====
Source 1
Distance: 0.5631
Content: Library Timings

Source 2
Distance: 0.5739
Content: Monday to Friday: 8:00 AM to 8:00 PM.
Saturday: 9:00 AM to 5:00 PM.
The library remains closed on Sundays and declared holidays.

Source 3
Distance: 1.1873
Content: Students can borrow up to 4 books at a time.
Books can normally be borrowed for 14 days.

===== GENERATED ANSWER =====
Saturday: 9:00 AM-5:00 PM.

Ln: 23 Col: 0
Ln: 172 Col: 0
```

Test Case 5 – Membership

User Query

How can I become a library member?

Retrieved Context

All currently enrolled students can become members of the college library by registering with their student ID.

Generated Output

Currently enrolled students can become library members by registering with their student ID.

Output:

```
CO3_AT2_Q18_Library_RAG.py - D:/CSA6502-Gen AI/CO3_AT2_Q18_Library_RAG.py (3.12.10)
File Edit Format Run Options Window Help

"""
Q18 - Library Information RAG System

Pipeline:
Document -> Chunking -> Embedding -> FAISS Retrieval
-> Hugging Face LLM -> Answer
"""

from sentence_transformers import SentenceTransformer
import faiss
from huggingface_hub import InferenceClient

# CONFIGURATION
# =====
HF_TOKEN = "hf_mWuSHDULcgkgZfRhFqMgIRSwWnRHHTXkn"
MODEL = "openai/gpt-oss-120b"
EMBEDDING_MODEL = "sentence-transformers/all-MiniLM-L6-v2"

# =====
# LIBRARY RULES DOCUMENT
# =====
LIBRARY_RULES = """
Library Membership

All currently enrolled students can become members of the
college library by registering with their student ID.

Borrowing Rules

Students can borrow up to 4 books at a time.
Books can normally be borrowed for 14 days.

Renewal Rules

A borrowed book can be renewed once for another 7 days,
provided that the book has not been reserved by another user.

Fines
"""

IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help

>>> Saturday: 9:00 AM-5:00 PM.

===== RESTART: D:/CSA6502-Gen AI/CO3_AT2_Q18_Library_RAG.py =====
LIBRARY INFORMATION RAG

Document chunks: 18
Loading embedding model...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
o enable higher rate limits and faster downloads.
Loading weights: 0% | 0/103 [00:00<?, ?it/s]Loading weights: 100% | ██████████
| 103/103 [00:00<00:00, 7358.43it/s]
FAISS vector database created.

Ask a library question: How can I become a library member?

Performing semantic search...

===== RETRIEVED CONTEXT =====
Source 1
Distance: 0.454
Content: Library Membership

Source 2
Distance: 0.749
Content: All currently enrolled students can become members of the
college library by registering with their student ID.

Source 3
Distance: 0.9204
Content: Registered library members may access available digital
library resources using their institutional credentials.

===== GENERATED ANSWER =====
Register with your student ID.

Ln: 41 Col: 0
Ln: 213 Col: 0
```

Test Case 6 – Unsupported Query

User Query

What is the hostel mess menu today?

Retrieved Context

The semantic search retrieves the closest available library-related information, but the required information about the hostel mess is not present in the knowledge base.

Expected Generated Output

The requested information is not available in the library rules.

The exact wording generated by the LLM may vary.

Output:

```
CO3_AT2_Q18_Library_RAG.py - D:/CSA6502-Gen AI/CO3_AT2_Q18_Library_RAG.py (3.12.10)
File Edit Format Run Options Window Help

Q18 - Library Information RAG System

Pipeline:
Document -> Chunking -> Embedding -> FAISS Retrieval
          -> Hugging Face LLM -> Answer

from sentence_transformers import SentenceTransformer
import faiss
from huggingface_hub import InferenceClient

# ===== CONFIGURATION =====
# CONFIGURATION
# =====

HF_TOKEN = "hf_mWuSHDulcgkgZfRhQMgIRSWWnHHHTXkn"
MODEL = "openai/gpt-oss-120b"
EMBEDDING_MODEL = "sentence-transformers/all-MiniLM-L6-v2"

# ===== LIBRARY RULES DOCUMENT =====
# LIBRARY RULES DOCUMENT
# =====

LIBRARY_RULES = """
Library Membership

All currently enrolled students can become members of the
college library by registering with their student ID.

Borrowing Rules

Students can borrow up to 4 books at a time.
Books can normally be borrowed for 14 days.

Renewal Rules

A borrowed book can be renewed once for another 7 days,
provided that the book has not been reserved by another user.

Fines
"""

IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help

>>>
===== RESTART: D:/CSA6502-Gen AI/CO3_AT2_Q18_Library_RAG.py =====
LIBRARY INFORMATION RAG

Document chunks: 18

Loading embedding model...
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN t
o enable higher rate limits and faster downloads.
Loading weights: 0% | 0/103 [00:00<?, ?it/s]Loading weights: 100% | ██████████
| 103/103 [00:00<00:00, 13792.65it/s]
FAISS vector database created.

Ask a library question: What is the hostel mess menu today?

Performing semantic search...

===== RETRIEVED CONTEXT =====

Source 1
Distance: 1.658
Content: Monday to Friday: 8:00 AM to 8:00 PM.
Saturday: 9:00 AM to 5:00 PM.
The library remains closed on Sundays and declared holidays.

Source 2
Distance: 1.6846
Content: Library Timings

Source 3
Distance: 1.7086
Content: Damaged Books

===== GENERATED ANSWER =====
The requested information is not available in the library rules.
```

Test Case Summary

Test Case	Query	Expected Behaviour	Status
TC1	How many books can I borrow?	Returns borrowing limit	PASS
TC2	Can I renew a borrowed book?	Returns renewal rule	PASS
TC3	What is the overdue fine?	Returns fine information	PASS
TC4	What are Saturday timings?	Returns Saturday timing	PASS
TC5	How can I become a member?	Returns membership rule	PASS
TC6	What is the hostel mess menu?	Identifies unavailable information	PASS

Error and Edge-Case Handling

The system handles several possible situations.

Empty Query

If the user enters no question:

[INPUT ERROR]

Question cannot be empty.

Irrelevant Query

For a question unrelated to the library:
The requested information is not available
in the library rules.

Unsupported Information

The system does not intentionally create a library rule that does not exist in the knowledge base.

API Error

If the Hugging Face API is unavailable or authentication fails, the program catches the exception and displays an API error.

Retrieved Context / Source Demonstration

One of the important requirements of the assessment is:
Display retrieved context/source wherever applicable.
The program satisfies this requirement by displaying:

RETRIEVED CONTEXT

Source 1

Distance: ...

Content:

Students can borrow up to 4 books at a time...

This allows the user and evaluator to see where the generated answer came from.

Results

The Library Information RAG system was successfully implemented using Python, Sentence Transformer embeddings, FAISS, and the Hugging Face LLM.

The system successfully answered questions related to borrowing, renewal, fines, timings, and membership.

The retrieved library context was displayed before the final answer was generated.

The system also handled unsupported questions by informing the user when the required information was not available in the library knowledge base.

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Pipeline Functionality	30%	Correct RAG pipeline — embeddings, retrieval and generation all function coherently	Pipeline mostly correct with minor gaps	Pipeline only partially functional	Pipeline non-functional or conceptually flawed
Answer Relevance & Groundedness	25%	Retrieves relevant context and generates accurate, grounded answers	Mostly relevant/accurate answers	Inconsistent relevance or accuracy	Answers largely irrelevant or hallucinated
Query Robustness	20%	Robust handling of varied queries and documents	Handles most queries well	Handles only simple queries	Fails on basic queries
End-to-End Demo	15%	Smooth, well-demonstrated end-to-end system	Demo mostly smooth, minor hiccups	Demo has noticeable issues	Demo fails or is not ready
Architecture Explanation	10%	Clear architecture diagram and explanation of design decisions	Adequate explanation of design	Minimal explanation of design	No explanation of design provided

Marks Evaluation:

Question No.	Pipeline Functionality(30%)	Answer Relevance & Groundedness (25%)	Query Robustness(20%)	End-to-End Demo(15%)	Architecture Explanation (10%)	Total Marks (100)
Q1						
Q2						
Overall Total (100)						